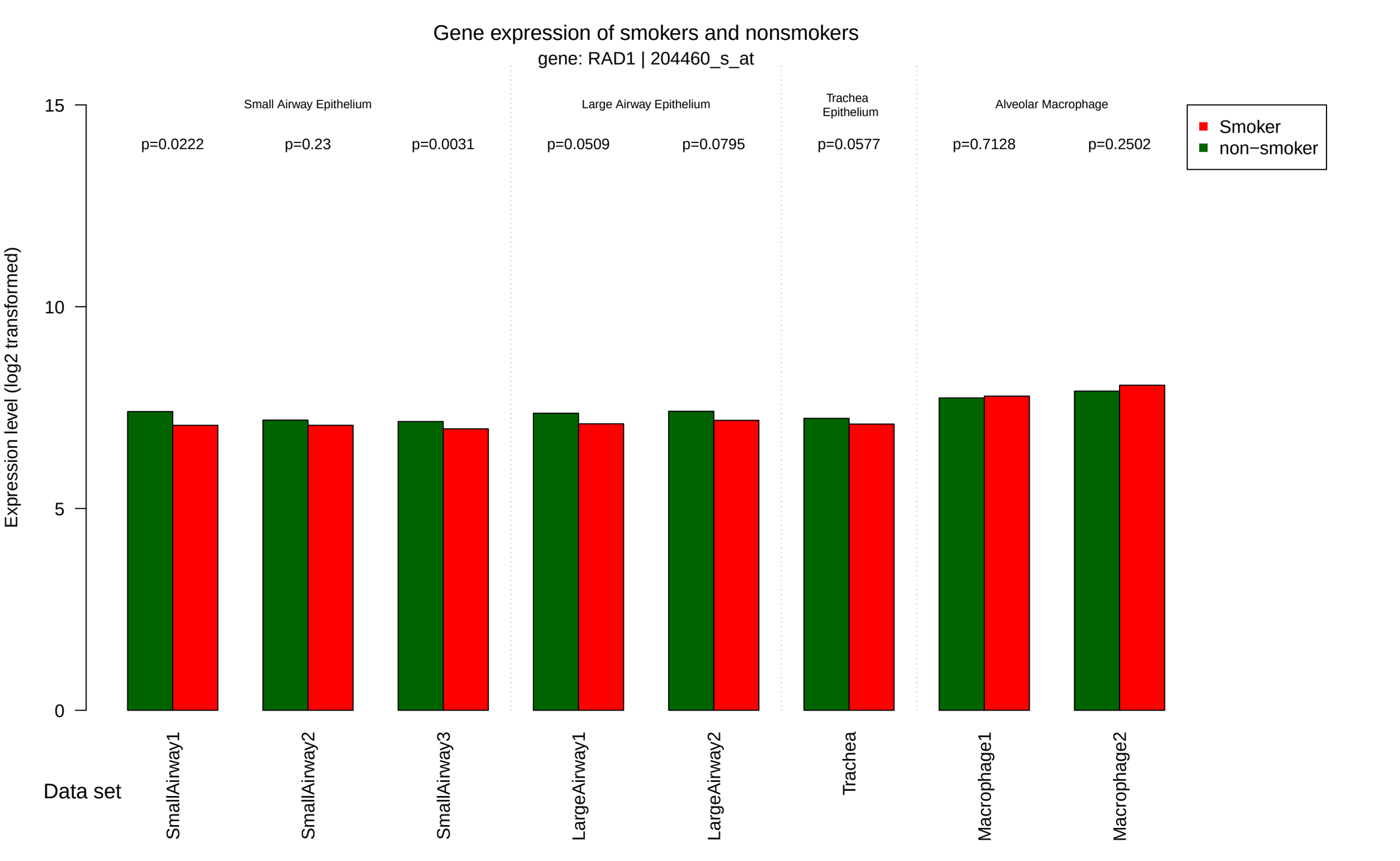
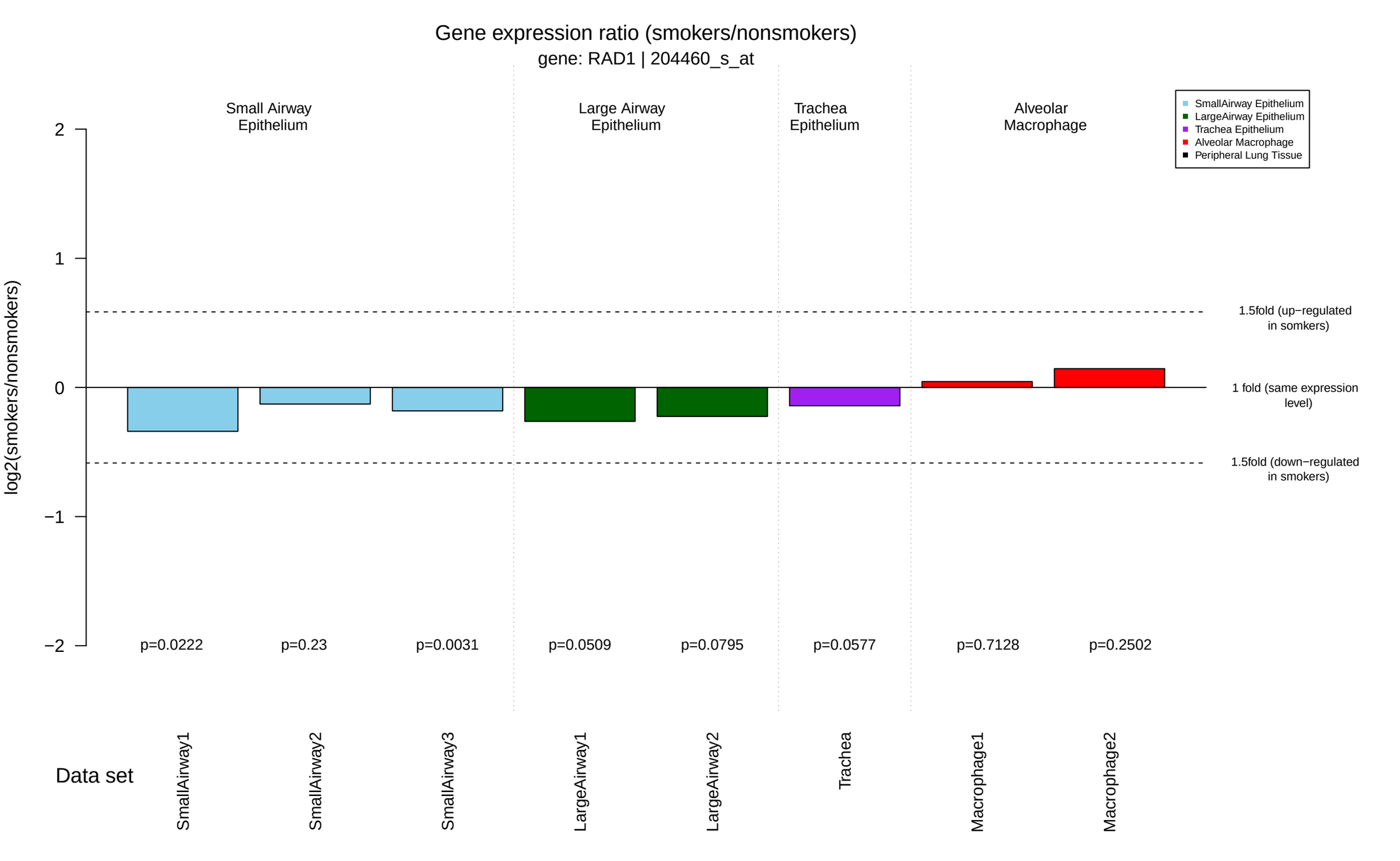
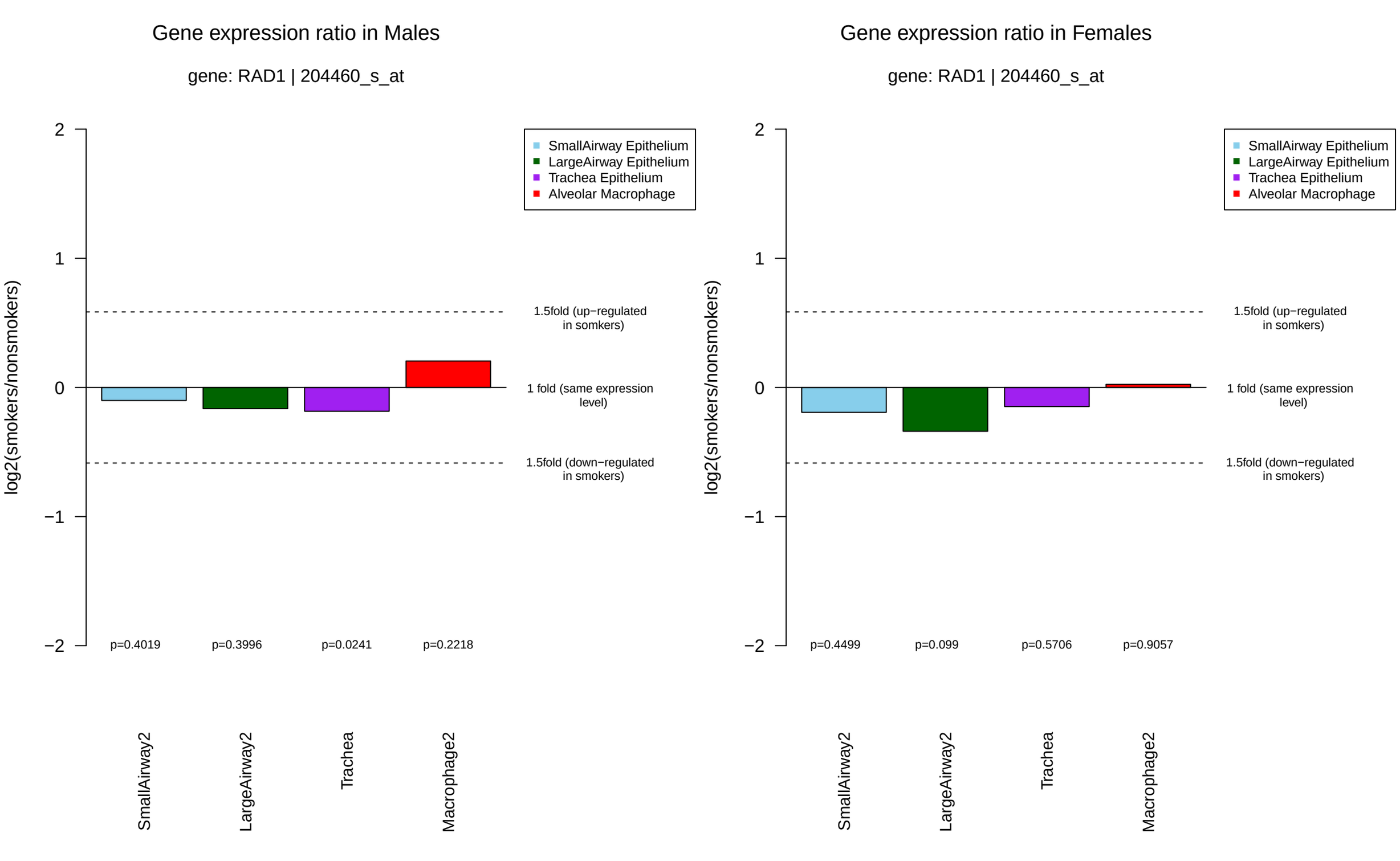
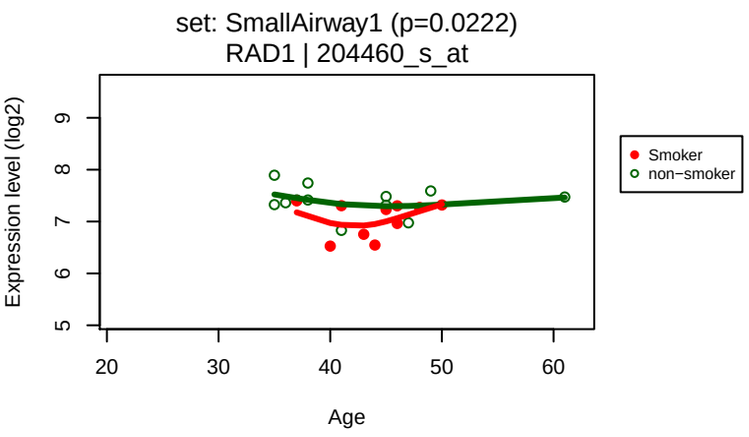
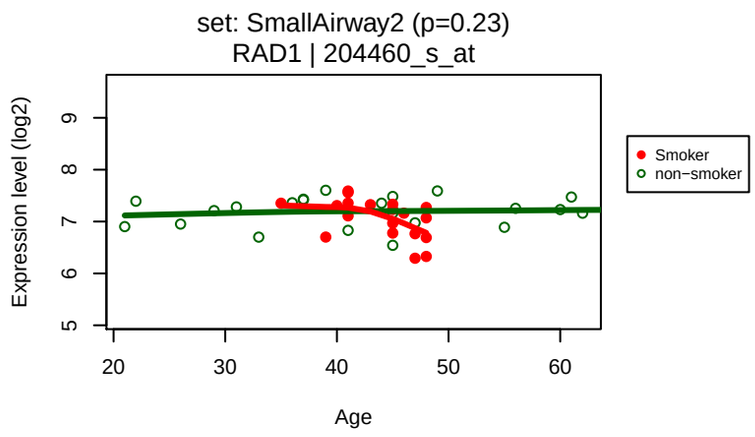
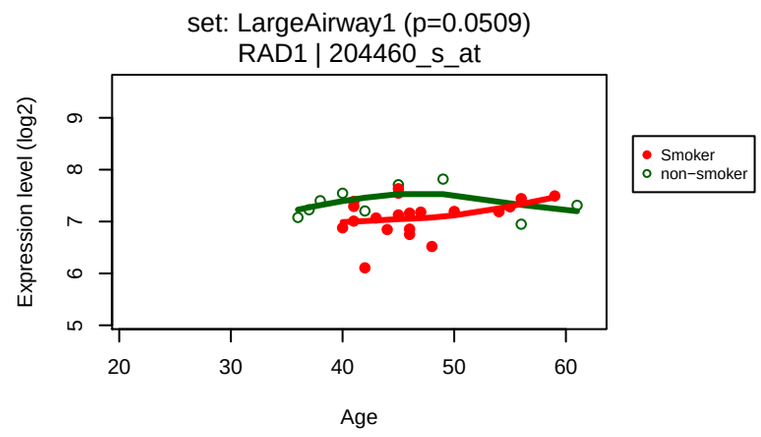

Gene expression of smokers and nonsmokers
gene: RAD1 | 204460_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0222
p=0.23
p=0.0031
p=0.0509
p=0.0795
p=0.0577
p=0.7128
p=0.2502
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

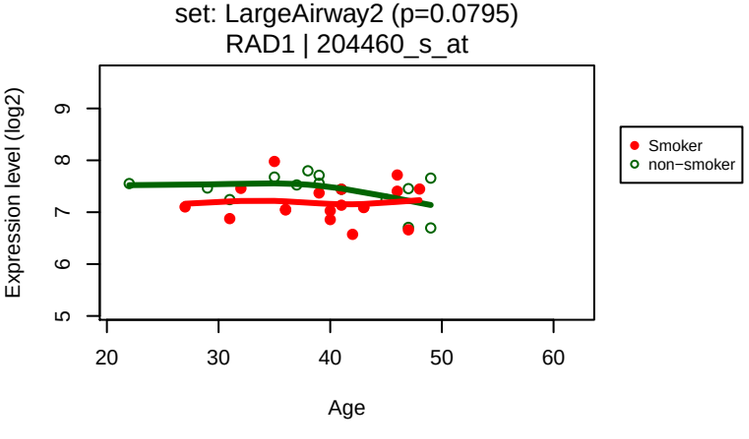

Gene expression ratio (smokers/nonsmokers)
gene: RAD1 | 204460_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0222
p=0.23
p=0.0031
p=0.0509
p=0.0795
p=0.0577
p=0.7128
p=0.2502
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

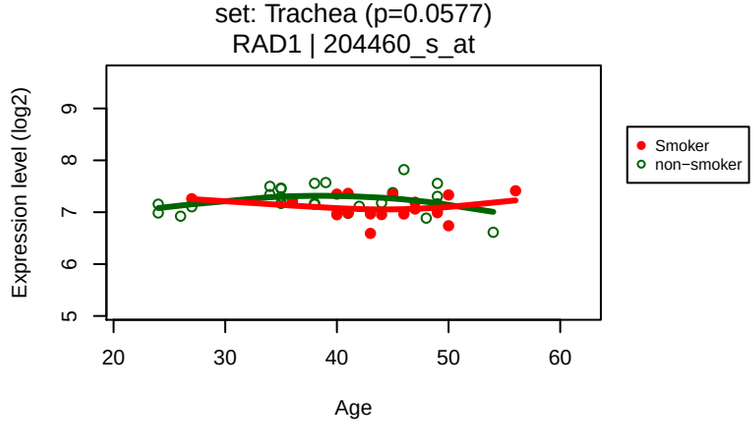

Gene expression ratio in Males
gene: RAD1 | 204460_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4019
p=0.3996
p=0.0241
p=0.2218
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD1 | 204460_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4499
p=0.099
p=0.5706
p=0.9057
SmallAirway2
LargeAirway2
Trachea
Macrophage2

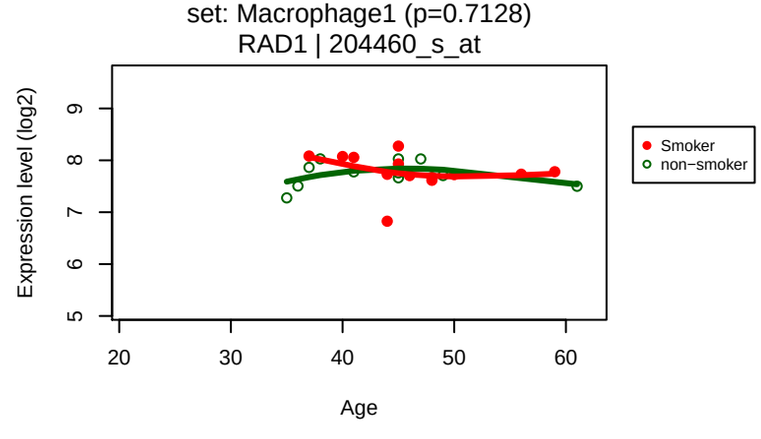

set: SmallAirway1 (p=0.0222)
RAD1 | 204460_s_at
Expression level (log2)
Age
Smoker
non−smoker

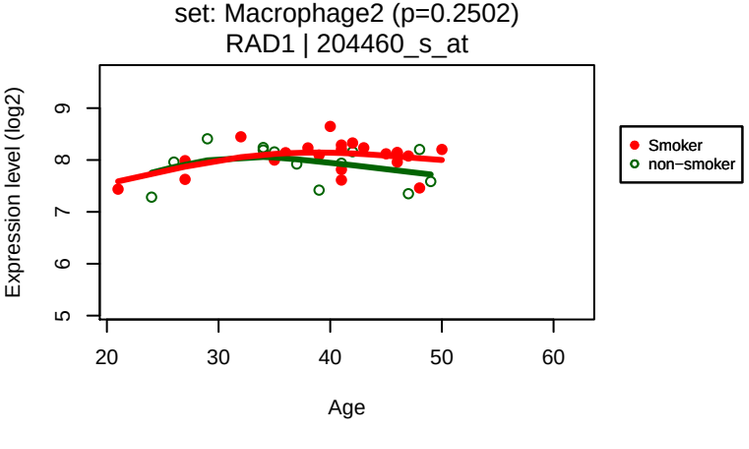

set: SmallAirway2 (p=0.23)
RAD1 | 204460_s_at
Expression level (log2)
Age
Smoker
non−smoker

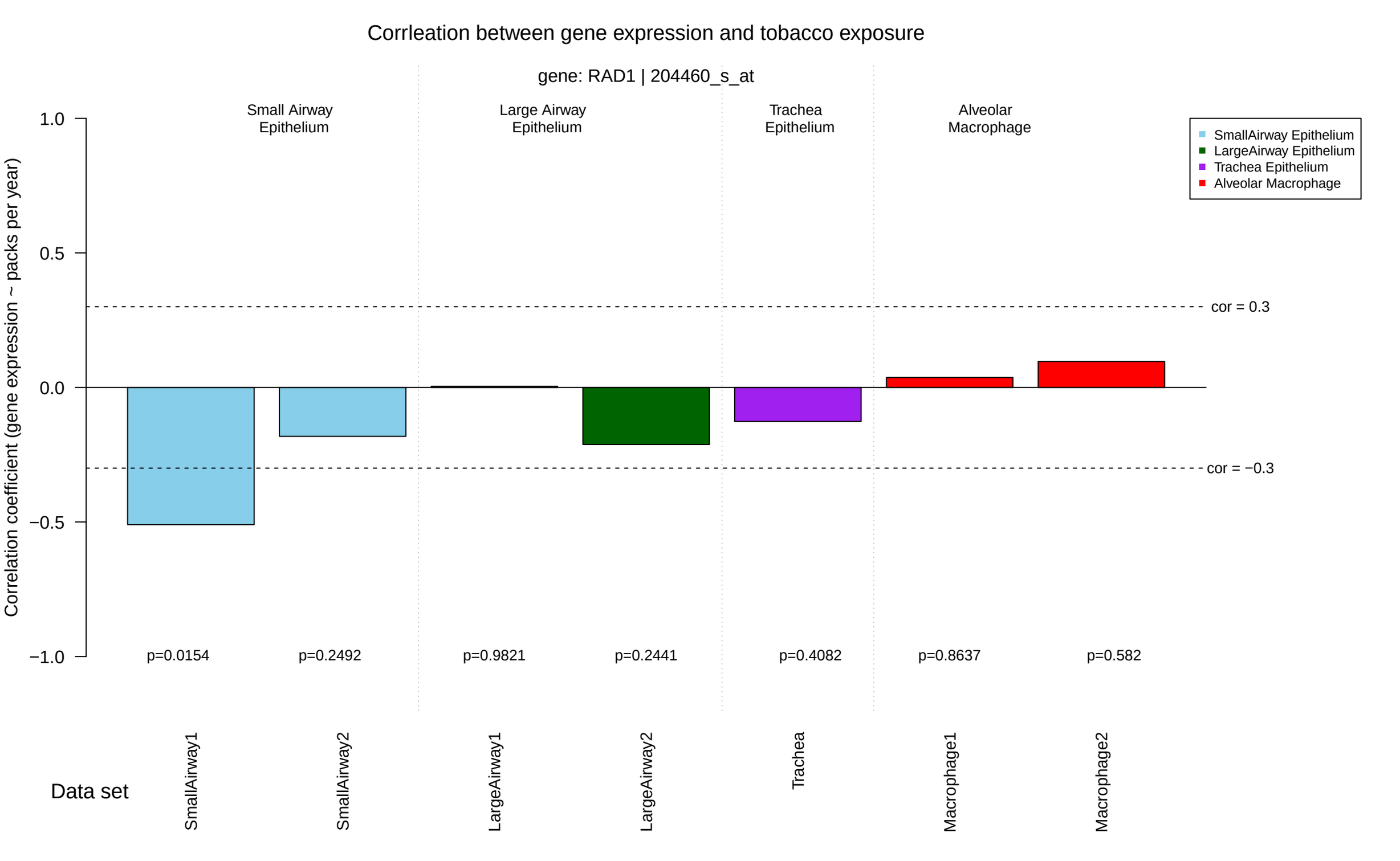

set: LargeAirway1 (p=0.0509)
RAD1 | 204460_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0795)
RAD1 | 204460_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0577)
RAD1 | 204460_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7128)
RAD1 | 204460_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2502)
RAD1 | 204460_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD1 | 204460_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0154
p=0.2492
p=0.9821
p=0.2441
p=0.4082
p=0.8637
p=0.582
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2